THE MODULUS INVERTS

The inverse of z has modulus : a number far from the origin has an inverse close to it, and vice versa. Comparing $|z|$ with 1 decides which.

MULTIPLICATIVE INVERSE

$$|z \quad| =$$

split by $|z|$ compared with 1

CASE 1

$|z| > 1$

Moves closer

the inverse has modulus less than one

E.G. $|z| = 5 \Rightarrow |z \quad| =$

CASE 2

$|z| < 1$

Moves farther out

the inverse has modulus greater than one

E.G. $|z| = \quad \Rightarrow |z \quad| = 3$

CASE 3

$|z| = 1$

Stays on the unit circle

the same distance from the origin after inversion

THE ARGUMENT NEGATES

In every case $\arg(z \quad) = -\arg(z)$: the inverse reflects z across the real axis and scales its distance by .